VitalSync: A Decoupled IoMT Architecture for Risk Triage and Resilient Clinical Summarization

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***Abstract*—Continuous monitoring in the Internet of Medical Things produces physiological telemetry far faster than clinicians can read it, while generative summarization offers relief at the cost of introducing a slow, unreliable and occasionally fabricating component into the critical path. We present VitalSync, an architecture that decouples high-rate telemetry ingestion and deterministic risk triage from generative summarization, keeping the system available through a provider cascade that terminates in a deterministic fallback. We make three contributions. First, we audit a widely used public vital-signs dataset and show that its risk label is a deterministic three-clause rule over heart rate and body mass index, exactly reproducing all 200,020 published labels; classifiers reach ceiling accuracy by recovering this Boolean, and an ablation on the remaining vital signs does not beat the majority-class baseline. Reported accuracies on this dataset are therefore artifacts of label construction. Second, we separate latent physiology from sensor observation and show that a learned triage model recovers the clean-signal NEWS2 escalation band more reliably than the NEWS2 rule applied to the same corrupted readings, with the advantage growing monotonically in sensor noise and vanishing—indeed reversing—when observations are noiseless. Third, we show that macro-F1 selects a model with 5.9× the rate of high-risk observations reported as low risk relative to a cost-aware criterion, and we report the disagreement rather than resolving it retrospectively. The implementation sustains 100 requests per second with sub-32 ms p95 ingestion latency, and continues to ingest and summarize with the triage service stopped. All results are software engineering evaluation on simulated cohorts; no clinical claim is made.**

***Index Terms*—Internet of Medical Things, remote patient monitoring, early warning score, clinical decision support, large language models, dataset auditing, system resilience**

I. Introduction

Remote patient monitoring generates continuous streams of vital-sign telemetry. The clinical difficulty is not acquisition but attention: a ward generating thousands of readings per hour presents clinicians with far more data than can be reviewed, and conventional per-parameter threshold alarms respond by generating notifications at a rate that erodes the value of each one.

Large language models are an appealing remedy, converting numeric streams into readable prose. They also introduce properties that a monitoring system should not inherit: latency measured in seconds rather than milliseconds, dependence on a remote service that may be rate-limited or unavailable, and a capacity to produce fluent text asserting things the data does not support. A system that places such a component in the path of telemetry ingestion inherits its worst case.

VitalSync separates these concerns. Ingestion terminates at a durable write; risk triage is a deterministic, millisecond-scale operation; generative summarization sits behind a cascade whose final tier is a template generator that cannot fail. The architecture's claim is availability: no single language model is required for the system to function.

In building it we found that the dataset the project was premised on could not support the claim it was intended to support. We report that finding as a primary contribution rather than working around it, because the dataset is public, widely downloaded, and its defect is invisible to the standard train-and-report workflow.

A. Contributions

1) A label-provenance audit of a public vital-signs dataset, recovering its exact generating rule and demonstrating by ablation that the vital signs it distributes carry no information about its published label.
2) An experimental design that separates latent physiology from sensor observation, making "learned model versus threshold rule" a falsifiable comparison rather than a tautology, together with the negative control that validates it.
3) Evidence that aggregate classification metrics select triage models that are worse by clinically asymmetric cost, quantified on both a standard cohort and a reduced sensor-only deployment.
4) An implementation and its measurements: decoupling verified under a stalled dependency, throughput to saturation, and full availability with the triage service stopped.

II. Related Work and Background

A. Remote monitoring and alarm burden

Reviews of continuous remote monitoring consistently identify data volume and notification burden as limiting factors rather than sensing accuracy. Systems that emit one notification per threshold breach per parameter scale notifications with the product of patients, parameters and sampling rate.

Research prototype. Not a medical device and not evaluated for clinical use.

B. Early warning scores

The National Early Warning Score 2 (NEWS2), standardized by the Royal College of Physicians, aggregates respiratory rate, oxygen saturation, supplemental oxygen, systolic blood pressure, pulse, consciousness and temperature into a single score with defined escalation bands. It is deterministic, auditable, and clinically established. We use it as a labeling function, not as a baseline to beat in the abstract—a distinction Section IV develops.

C. Interoperability and regulatory context

HL7 FHIR represents vital signs as `Observation` resources, which shapes the data contract used here. The NIST AI Risk Management Framework and its generative AI profile provide lifecycle risk guidance, and FDA guidance on Clinical Decision Support Software delimits when such a system becomes a regulated device. These are cited as context for scope, never as evidence of efficacy: this work makes no clinical claim and the prototype is not a device.

D. Language models over clinical text

Reviews of clinical text summarization with language models repeatedly identify hallucination, evaluation methodology, bias and privacy as unresolved. Our design responds to the first directly: generated output is mechanically checked against its inputs before it is accepted, and rejected output causes the cascade to fall through rather than being surfaced.

III. Track A: Auditing the Dataset

A. Motivation

The project was specified around a public Kaggle dataset of human vital signs (200,020 rows, 17 columns) with a `Risk Category` label, to be classified with random forest and gradient boosting. Before training we audited the label.

B. Method

We computed per-class descriptive statistics for every numeric column, ranked features by mutual information with the label, fit a depth-3 decision tree and exported its rule, tested a candidate Boolean against all rows, reproduced the specified random forest and gradient boosting experiment, and ran an ablation restricted to vital signs that showed no class separation.

C. Result: the label is a rule

The published label is reproduced exactly, with zero mismatches across all 200,020 rows, by

$$\text{High} \iff (\text{HR} > 90) \vee (\text{BMI} > 30) \vee (\text{BMI} < 18.5). \quad (1)$$

Three consequences follow. The dataset is binary; the three-class target originally specified cannot be trained from it. Every vital sign lies inside its normal reference range—heart rate 60–99, oxygen saturation 95.0–100.0, systolic pressure 110–139, temperature 36.0–37.5, respiratory rate 12–19—so the cohort contains no physiologically abnormal observation. Class-conditional means for oxygen saturation, both blood pressures, temperature, respiratory rate and mean arterial pressure agree to three decimal places.

TABLE I

Track A. Classification on the public vital-signs dataset, and the recovered label rule. Scores approach the ceiling because the published label is a deterministic function of heart rate and BMI, not because vital signs predict risk.

Model	Accuracy	Macro-F1
random_forest	1.00000	1.00000
gradient_boosting	1.00000	1.00000
Recovered rule	1.00000	–
Vitals-only ablation	0.51850	0.42085
Majority-class baseline	0.52552	–

Rule: `(Heart Rate > 90) OR (Derived_BMI > 30) OR (Derived_BMI < 18.5)`, exact on 200,020 of 200,020 rows (0 mismatches). The ablation uses only SpO_2, blood pressure, temperature and respiratory rate, and does not beat the majority-class baseline.

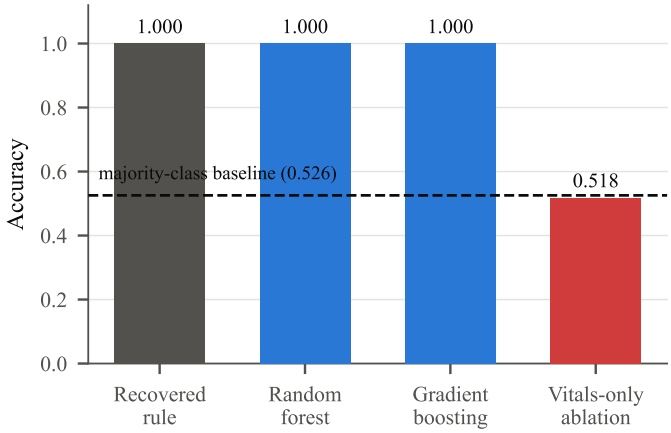


Fig. 1. Track A. Both ensembles reach the ceiling by recovering the label rule. The ablation trained only on oxygen saturation, blood pressure, temperature and respiratory rate falls *below* the majority-class baseline: those channels carry no information about the published label.

Table I reports the classification results alongside the rule and the ablation. Both ensembles reach the ceiling. The ablation, trained only on oxygen saturation, blood pressure, temperature and respiratory rate, does not beat the majority-class baseline: those five channels carry no information about the label.

D. Implication

A classification score on this dataset measures recovery of Equation 1. It is not evidence of clinical predictive capability, and reporting it as such would not survive a reviewer fitting a shallow tree. We therefore do not use this dataset to train the deployed model.

IV. Track B: Triage Under Sensor Noise

A. The circularity problem

Substituting NEWS2 for the dataset's label does not by itself solve anything. NEWS2 is also a deterministic function of vital signs, so a model trained to reproduce NEWS2 from

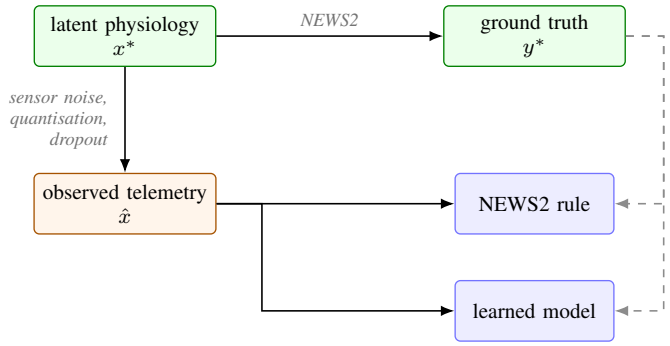


Fig. 2. Separating latent physiology from sensor observation. Ground truth is scored on the latent signal; every method, including the rule baseline, observes only the corrupted telemetry; the dashed edges indicate that both are scored against the same latent-signal truth. This is what makes "learned model versus threshold rule" a real comparison rather than a restatement of the labelling function.

clean vitals is learning a rule that is already known, and any reported accuracy is a statement about function approximation.

B. Design: latent truth versus observation

We break the circularity by distinguishing what is true from what is measured:

$$x^* \xrightarrow{\text{NEWS2}} y^*, \qquad x^* \xrightarrow{\text{sensor noise}} \hat{x},$$

where x^* is latent physiology, y^* the ground-truth escalation band, and $\hat{x}$ the corrupted telemetry. Ground truth is scored on x^*. Every method—including the rule baseline—observes only $\hat{x}$. The rule baseline computes NEWS2 on $\hat{x}$, which is precisely what a conventional threshold monitor does. The learned model is trained on $(\hat{x}, y^*)$ and can therefore internalize the noise distribution and exploit cross-parameter redundancy that a per-parameter rule cannot.

This yields a falsifiable comparison with a built-in negative control: at zero noise $\hat{x} = x^*$, the rule is exact by construction, and the learned model *must not* beat it. Fig. 2 shows the arrangement.

C. Cohort and corruption model

Latent severity is drawn from a three-component mixture (60% stable, 25% mild derangement, 15% acute deterioration) with four deterioration archetypes, so channels move together as they do clinically rather than varying independently. Corruption is applied per channel: Gaussian noise on all channels, motion artifact on heart rate, perfusion dropout on oxygen saturation, display quantization, and 3% channel dropout imputed at inference. Train and test are split by patient, never by row, since repeated readings from one simulated patient are correlated.

D. Metrics

Macro-F1 is the pre-registered selection criterion. It is insufficient alone: it scores a missed high-risk observation identically to a spurious one. We therefore also report high-risk recall, *missed-HIGH* (the fraction of high-risk observations

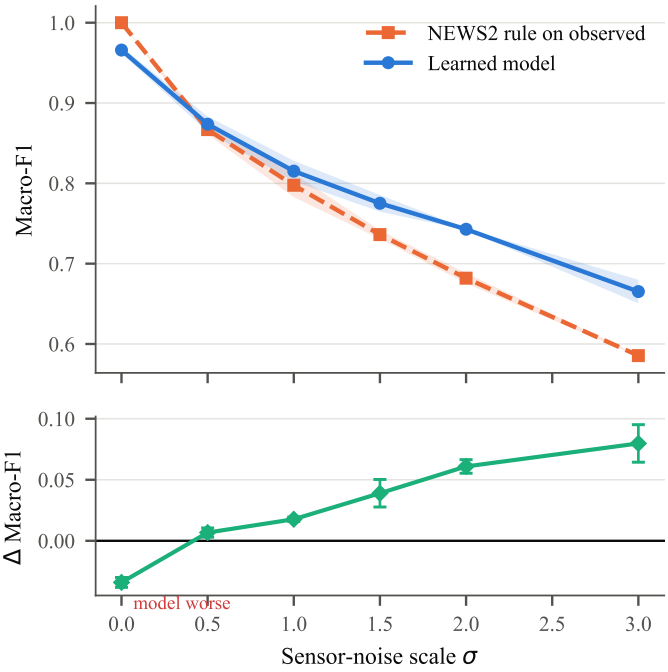


Fig. 3. Macro-F1 against sensor-noise scale (top) and the model-minus-rule difference with 95% confidence intervals over three cohort seeds (bottom). Shaded bands are 95% intervals. The difference changes sign: at $\sigma = 0$ the rule is exact and the learned model is significantly worse, and the advantage appears only once observations are corrupted.

reported as low risk), and a mean cost under an asymmetric matrix penalizing under-triage far more heavily than over-triage. Those weights are a stated assumption, not derived from outcome data.

E. Results

Table II reports all five methods on the same corrupted observations. Table III sweeps the noise scale, averaging three independent cohort seeds per cell and reporting a 95% confidence interval on the model-minus-rule difference.

The negative control holds: at $\sigma = 0$ the rule is exact and the learned model is significantly *worse*. The advantage appears once observations are corrupted and grows monotonically with noise, every interval excluding zero. This is the paper's central empirical claim, and its credibility rests on the fact that the design permits it to fail.

F. Selection criteria disagree

Macro-F1 and the cost criterion select different models. The F1-optimal model attains higher aggregate scores while reporting high-risk observations as low risk 5.9× more often than the cost-optimal alternative, for a macro-F1 difference of less than 0.006. We report the disagreement and retain the pre-registered criterion; changing it after observing results would be precisely the post-hoc selection that makes a reported number untrustworthy.

TABLE II

TRACK B. RECOVERING THE CLEAN-SIGNAL NEWS2 BAND FROM SENSOR-CORRUPTED TELEMETRY. ALL ROWS SEE THE SAME CORRUPTED OBSERVATIONS. MISSED-HIGH IS THE FRACTION OF HIGH OBSERVATIONS REPORTED AS LOW.

Model	Accuracy	Macro-F1	HIGH recall	Missed-HIGH	Mean cost
news2_rule_on_observed	0.9005	0.8006	0.8847	0.0127	0.3165
threshold_alarms	0.7352	0.6280	0.8520	0.0070	0.4461
logistic_regression	0.8677	0.7866	0.8398	0.0023	0.2893
random_forest	0.9005	0.8267	0.9013	0.0014	0.2074
gradient_boosting	0.9260	0.8324	0.9199	0.0082	0.2651

Cohort: 120,000 readings from 30,000 simulated patients, split by patient. Selection metric: `macro_f1`. Macro-F1 selects `gradient_boosting` while the asymmetric-cost criterion selects `random_forest`; the F1-optimal model carries the higher rate of HIGH observations reported as LOW. Cost weights are a stated assumption, not derived from outcome data.

TABLE III

MACRO-F1 AGAINST SENSOR-NOISE SCALE, MEAN OVER 3 INDEPENDENT COHORT SEEDS. AT $\sigma = 0$ THE OBSERVATION EQUALS THE LATENT SIGNAL, SO THE RULE IS EXACT AND THE LEARNED MODEL CANNOT IMPROVE ON IT; THE ADVANTAGE APPEARS ONLY ONCE OBSERVATIONS ARE CORRUPTED.

σ	NEWS2 rule	gradient_boosting	Δ	95% CI
0.0	1.0000	0.9659	-0.0341	[-0.0381, -0.0300]
0.5	0.8669	0.8737	+0.0067	[+0.0030, +0.0105]
1.0	0.7976	0.8152	+0.0177	[+0.0158, +0.0195]
1.5	0.7362	0.7751	+0.0390	[+0.0277, +0.0502]
2.0	0.6819	0.7428	+0.0609	[+0.0553, +0.0665]
3.0	0.5855	0.6653	+0.0798	[+0.0644, +0.0951]

† interval includes zero; the difference is not resolved at this number of seeds.

TABLE IV

CALIBRATION OF THE SELECTED TRIAGE MODEL. A POSITIVE GAP MEANS STATED CONFIDENCE EXCEEDS OBSERVED ACCURACY IN THAT BIN.

Confidence bin	n	Mean conf.	Accuracy	Gap
[0.3, 0.4]	26	0.387	0.500	-0.113
[0.4, 0.5]	414	0.471	0.454	+0.016
[0.5, 0.6]	1,054	0.550	0.548	+0.001
[0.6, 0.7]	1,063	0.651	0.611	+0.039
[0.7, 0.8]	1,009	0.751	0.733	+0.018
[0.8, 0.9]	1,283	0.853	0.843	+0.010
[0.9, 1.0]	19,151	0.988	0.991	-0.002

ECE 0.0053, MCE 0.1131, Brier 0.1022, overall over-confidence +0.0014.

G. Calibration

The system displays a confidence value, so whether that value means anything is part of the result rather than an implementation detail. Table IV gives the reliability table.

H. Sensor-only deployment

Two NEWS2 parameters—supplemental oxygen and consciousness—are recorded by staff, not sensed by a wearable. A deployment fed only by wearables cannot supply them, and the interface defaults them to the benign case. Table V quantifies the resulting silent train/serve mismatch and shows that retraining for the inputs actually present recovers most of the safety margin while barely changing macro-F1—another instance of the aggregate metric concealing the clinically relevant difference.

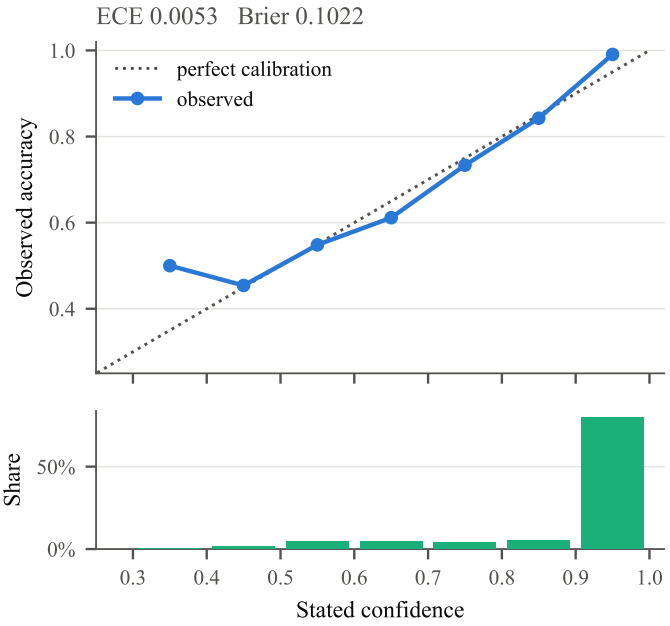


Fig. 4. Reliability diagram for the selected triage model, with bin support below. Points on the diagonal indicate that a stated confidence matches observed accuracy. Support is shown because a large gap in a sparsely populated bin is a weaker finding than the same gap in a well populated one.

TABLE V

WEARABLE-ONLY DEPLOYMENT ABLATION. SUPPLEMENTAL OXYGEN AND AVPU ARE NEWS2 PARAMETERS THAT A WEARABLE CANNOT SUPPLY.

Scenario	Macro-F1	HIGH recall	Missed-HIGH
Full observation (reported setting)	0.8214	0.8891	0.0051
Trained full, served wearable-only	0.7783	0.7803	0.0167
Trained and served wearable-only	0.7744	0.8448	0.0109

Serving the fully-trained model on a wearable-only feed costs +0.0431 macro-F1 and raises missed-HIGH by +0.0116.

V. SYSTEM ARCHITECTURE

A. Overview

Four components: a Python triage service exposing a stateless inference API; a Java backend owning authentication, validation, persistence, orchestration and the summarization cascade; PostgreSQL for durable telemetry and assessments;

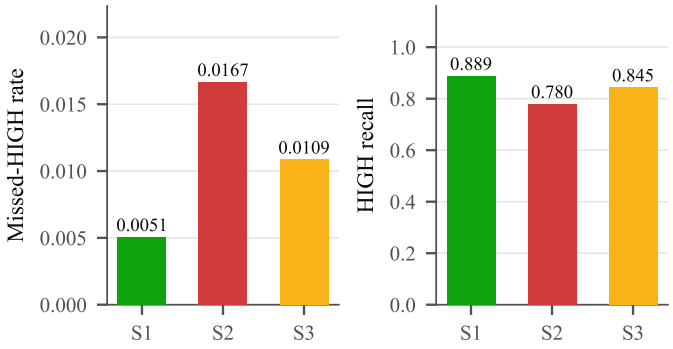


Fig. 5. Sensor-only deployment. Serving the fully-trained model against a feed lacking supplemental-oxygen and consciousness observations (S2) raises the rate of high-risk observations reported as low risk more than three-fold. Retraining for the inputs actually present (S3) recovers most of the safety margin while barely changing macro-F1.

and a browser dashboard. Fig. 6 shows the arrangement and, more importantly, the boundary between what must be fast and what is permitted to be slow.

B. The decoupling rule

Telemetry ingestion performs validation and one durable write, then returns `202 Accepted`. Analysis is handed to a bounded executor with a caller-runs rejection policy, so sustained overload applies backpressure to ingestion instead of growing an unbounded queue of pending model calls. Ingestion throughput is bounded by the database, never by whichever language model is currently answering.

C. Summarization cascade

Generation is attempted best-first: a local model, then a cloud model, then a deterministic template generator with no external dependency. Each tier may fail or time out; the terminal tier cannot, which is what makes the cascade total. The tier that answered and its latency are persisted per assessment, so resilience is measured from stored data rather than inferred from logs. Fig. 7 shows the flow, including the faithfulness gate described below.

D. Faithfulness enforcement

Prompt instructions are not enforcement. Generated text is mechanically checked before acceptance: every number it cites must correspond to a supplied value within a tolerance that admits rounding but not fabrication, and no named condition may appear. Output failing either check is treated as a tier failure and the cascade falls through. Deterministic output is checked by the same guard and is verified in test to always pass.

This is necessary, not sufficient. It catches invented values; it cannot detect a fluent, numerically correct, clinically wrong interpretation.

TABLE VI
TRIAGE INFERENCE LATENCY, 1000 SINGLE-OBSERVATION PREDICTIONS. IN-PROCESS MODEL CALL ONLY: EXCLUDES HTTP, SERIALISATION AND NETWORK TRANSIT.

Statistic	Milliseconds
Mean	2.758
p50	2.684
p95	3.719
p99	4.297
Maximum	4.687

Measured on one developer workstation. Not a service-level objective.

VI. System Evaluation

A. Triage latency

Triage is the only model-dependent step in the synchronous-facing path, so its cost bounds how much of the ingestion budget analysis can consume. Table VI reports the in-process model call over 1000 single-observation predictions. The distribution is tight and the tail is short, which is what permits analysis to be scheduled per reading rather than batched.

B. Decoupling under a stalled dependency

With the triage service artificially stalled for 3 s, ingestion returns in under 1 s. A synchronous implementation cannot pass this test, which is why it is the load-bearing test of the architectural claim.

C. Throughput

Offered load was generated open-loop, so a server falling behind manifests as rising latency rather than a silently reduced offered rate. At 10, 50 and 100 requests per second the system tracks the offered rate with p95 ingestion latency between 27 and 31 ms. At an offered 250 it saturates, achieving approximately 159 requests per second with p95 latency rising to several seconds. The asynchronous analysis backlog drained completely at every rate. Fig. 8 shows both panels; achieved rate and latency are plotted separately rather than on twin axes, since they share no scale.

D. Resilience

Three conditions were exercised by stopping and restarting containers. With all services running, every reading produced an assessment. With the triage service stopped, ingestion and summarization both remained at 100%, assessments being recorded with the model marked unavailable and summaries produced by the deterministic tier. On restart the system resumed using the model without a backend restart.

E. Notification event counts

We stress that this is arithmetic over a simulated stream. It is not a measurement of alert fatigue, clinician workload, or patient benefit, and it must not be read as one. A policy emitting fewer notifications is not thereby safer; fewer notifications can mean missed deterioration.

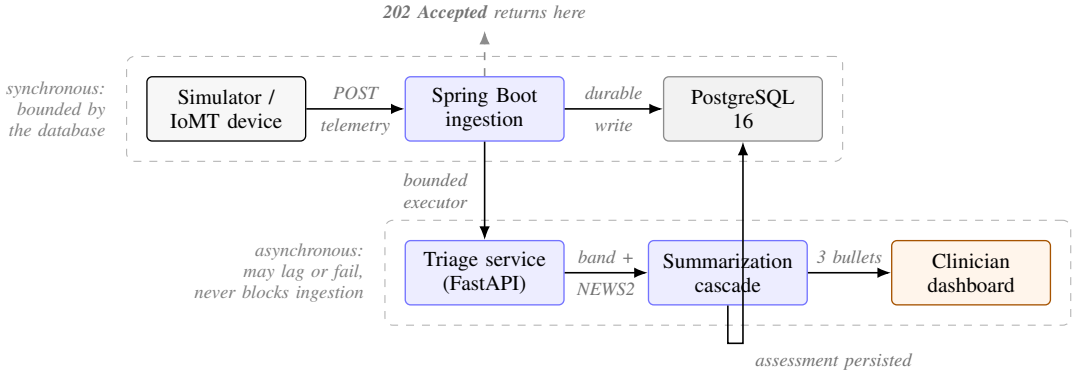


Fig. 6. Architecture and the synchronous/asynchronous boundary. Ingestion performs one durable write and returns `202 Accepted`; triage and summarization proceed on a bounded executor behind it. Ingestion throughput is therefore bounded by the database, never by whichever language model is currently answering.

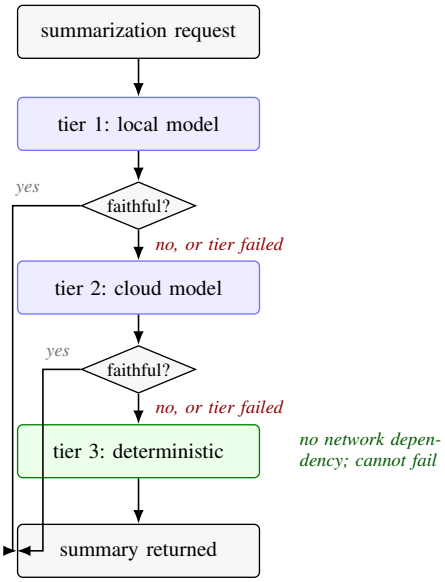


Fig. 7. Summarization cascade. A tier that fails, times out, or returns output failing the faithfulness check is skipped. The deterministic tier has no external dependency and therefore always produces a summary, which is the property the availability claim rests on.

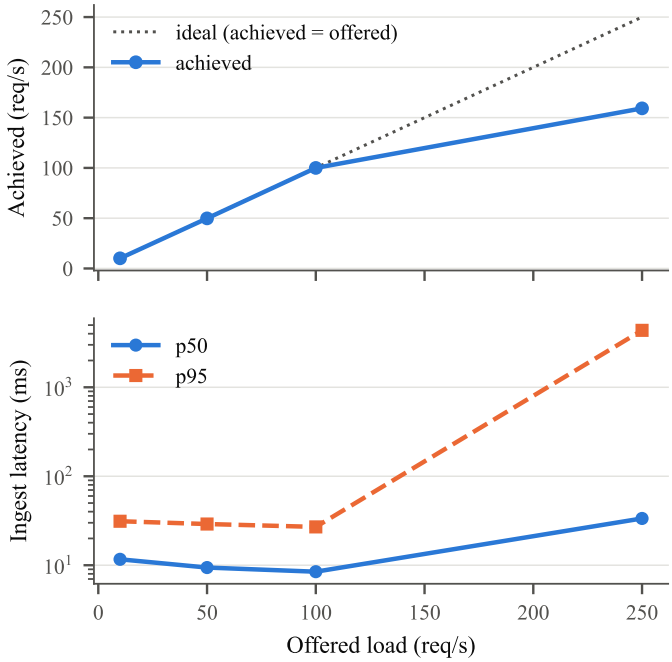


Fig. 8. Ingestion under open-loop offered load. Achieved rate tracks offered load to 100 requests per second and then departs from it; p95 latency rises by two orders of magnitude at the saturation point. Latency uses a logarithmic axis.

TABLE VII
NOTIFICATION EVENT COUNTS OVER 4,800 SIMULATED READINGS. A EMITS ONE EVENT PER BREACHING PARAMETER PER READING; B ONE PER READING WHOSE BAND IS NOT LOW; C ONE PER UPWARD BAND TRANSITION PER PATIENT. THIS IS EVENT ARITHMETIC ONLY.

Policy	Events	Per 100
A. Per-parameter thresholds	4,306	89.71
B. Aggregate band	1,262	26.29
C. Band transition	419	8.73

These counts are not a measurement of alert fatigue, clinician workload, or patient benefit. Policy C emits fewer notifications by construction; whether that is clinically safe is untested here.

VII. Live Language Model Evaluation

This section is reserved for evaluation of the live summarization tiers and is deliberately unpopulated in the present version. The cascade, its failure paths and the faithfulness guard are implemented and tested, but no language model has been executed: every summary reported in this paper was produced by the deterministic tier. Consequently the faithfulness guard has been shown to reject fabricated output only under test, and its behaviour against real model output is unmeasured.

We regard reporting this as an unmeasured gap as preferable to omitting the section. The availability argument of Section V rests on tiers whose generative behaviour has not yet been characterized, and that limit should be visible where a reader looks for those results.

VIII. Discussion

A. On auditing before modeling

The Track A finding was not subtle and required no sophisticated tooling—a depth-3 tree exposes it in seconds. It survives in the literature because the standard workflow never asks the question. We suggest that any classification result on a public physiological dataset should be accompanied by evidence that the label is not a deterministic function of the features, and that this is cheap enough to be routine.

B. On negative controls

The $\sigma = 0$ result, where the learned model is significantly worse than the rule, is the most informative number in Table III. A design in which the model wins everywhere would be indistinguishable from a design that leaks.

C. On aggregate metrics in clinical triage

Macro-F1 selected the worse model twice, under different circumstances: once between estimator families, once between input configurations. In both cases the cost-relevant difference was invisible in the aggregate score. This is an argument for reporting directional error rates alongside aggregate metrics whenever error costs are asymmetric.

IX. Limitations

1) The Track B cohort is simulated. Results are software evaluation, not clinical validation, and the corruption model is a stated assumption.
2) No language model has been run live; Section VII is unpopulated, and the faithfulness guard is unvalidated against real generative output.
3) Faithfulness checking cannot detect fluent, numerically correct, clinically incorrect interpretation.
4) The asymmetric cost weights encode intuition, not outcome data.
5) NEWS2 is validated for acutely ill adults in hospital; applying it to simulated continuous wearable telemetry is outside its validation scope.
6) Security is prototype grade: a single demo credential, no refresh or revocation, no external identity provider, no TLS in the bundled stack.
7) Throughput figures characterize one workstation and are not service level objectives.

X. Ethics and Safety

The system is a research prototype and not a medical device. It produces software triage categories, not diagnoses, and every response carries that statement. No real patient data was used at any stage; all cohorts are simulated and the audited public dataset contains no identifiable information. Enabling the cloud summarization tier transmits telemetry to a third party, which is acceptable for synthetic data and would require a data processing agreement otherwise. Deployment on real data would require the hardening enumerated under Limitations, and a determination of regulatory status under applicable Clinical Decision Support guidance.

XI. Reproducibility

All results are regenerated from a clean checkout by `run_all.sh` or `run_all.ps1`. Each experiment writes a JSON artifact recording its seeds, configuration and environment; every table in this paper is generated from those artifacts by `paper/make_tables.py`, and no number is entered by hand. The audited dataset is not redistributed; a resolver script locates it and records its SHA-256 so a result can be tied to a specific file. Track B requires no external data.

XII. Conclusion

VitalSync demonstrates that continuous telemetry ingestion and low-latency risk triage can be isolated from generative summarization, remaining fully available when every language model tier is unreachable. Along the way we found that the public dataset the work was premised on carries a label that is a deterministic rule over two variables, and we report that as a caution about label provenance. Separating latent physiology from sensor observation gives a falsifiable comparison in which a learned model outperforms a threshold rule only under observation noise—and demonstrably fails to when there is none. Finally, we show that aggregate classification metrics can select triage models that are materially worse under asymmetric clinical cost. Evaluating the generative tiers themselves remains open, and is marked as such.

Acknowledgment

The authors used an AI coding assistant for implementation and analysis scaffolding. All reported measurements were produced by executing the referenced code, and the artifacts are included.

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[7] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.

[8] N. Ayub, "Human Vital Sign Dataset," Kaggle, 2024. https://www.kaggle.com/datasets/nasirayub2/human-vital-sign-dataset. Audited in Section III; its risk label is a deterministic rule.